Is the Band Worth It? A Multi-Seed Comparison of RMSNorm Variants

An empirical evaluation of BandRMS, ZeroCenteredRMSNorm, and ZeroCenteredBandRMSNorm across five regimes

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Abstract

We empirically evaluate three variants of RMSNorm normalization — BandRMS (output magnitude constrained to a learnable band $[1-\alpha, 1]$), ZeroCenteredRMSNorm (mean subtracted before RMS rescaling), and ZeroCenteredBandRMSNorm (both modifications combined) — against vanilla RMSNorm across five settings: ViT-pico on CIFAR-10, ResNet-18 on CIFAR-100, a 4-layer pre-norm transformer on IMDb, a 24-block deep residual stack under mixed precision, and a layer-level synthetic microbenchmark. We run n=5 random seeds per cell and report mean ± std, 95% bootstrap CI, and a paired sign-flip permutation p-value (B=10,000, Phipson–Smyth corrected) against the RMSNorm baseline. Four probes log per-epoch activation mean, per-sample RMS distribution, internal scale state, and gradient norm at every normalization layer, testing each variant's specific theoretical claim independently of accuracy. Three findings emerge. First, **the band constraint by itself produces no measurable benefit on any task we tested**: BandRMS mechanistically confines output RMS to $[0.9, 1.0]$ as advertised, but yields zero accuracy or loss improvement over RMSNorm across all five experiments. Second, **zero-centering is the operative innovation**: ZeroCenteredRMSNorm produces near-zero output mean ($|\mathbb{E}[x]|$ reduced 30–500×) and translates this into a decisive +41.8 pp swing on ResNet/CIFAR-100 (random → learning), a −28% validation-loss reduction with 2.8× lower seed variance on the deep-fp16 stack, and modest lifts elsewhere. Third, **combining both gives the tightest variance** (62% lower std on ViT/CIFAR-10) and the best accuracy on the two vision tasks, but the band's tightness slightly hurts on the 24-block deep stack. We document a cross-experiment dissociation: shallow pre-norm transformers are mechanistically insensitive to norm-variant choice (E3 universal null within ±0.1 pp), while deep or post-norm-like regimes are where residual-stream DC drift dominates and zero-centering becomes critical.

1 Introduction

The default normalization layer in modern transformers is LayerNorm [Ba et al., 2016]:

$$x \;\mapsto\; \gamma \odot \frac{x - \mu(x)}{\sigma(x)} \;+\; \beta,$$

where $\mu(x)$ and $\sigma(x)$ are the mean and standard deviation along a designated axis and γ, β are learnable per-feature gain and bias. RMSNorm [Zhang & Sennrich, 2019] drops the mean subtraction and the bias, computing only

$$x \;\mapsto\; \gamma \odot \frac{x}{\sqrt{\mathbb{E}[x^2] + \epsilon}}$$

— about 1.5–2× faster in practice, and the dominant choice in LLaMA, Mistral, Qwen, and most contemporary large language models.

We study three extensions to RMSNorm:

- **BandRMS** (proposed here) replaces the per-feature gain $\gamma \in \mathbb{R}^d$ with a single learnable scalar $\beta \in \mathbb{R}$ whose effective scale is constrained to a band $[1-\alpha, 1]$ via a sigmoid. The motivation is that the rigid unit sphere of RMSNorm may be unnecessarily restrictive; a "thick spherical shell" (a learnable $(1-\alpha)$-thick neighborhood of the unit sphere) was hypothesized to give richer representations.

- **ZeroCenteredRMSNorm**, recently introduced as a design choice in Qwen3-Next [Qwen Team, 2025], re-introduces the mean subtraction of LayerNorm while keeping the per-feature gain of RMSNorm. The motivation is that in deep residual networks the residual stream accumulates a DC component (a non-zero mean) across layers, forcing γ to grow compensatorily; subtracting the mean before normalization removes this driver.

- **ZeroCenteredBandRMSNorm** (proposed here) composes the two: subtract mean, rescale by RMS of the centered tensor, multiply by the band-constrained scalar scale.

BandRMS and ZeroCenteredBandRMSNorm are our own design. ZeroCenteredRMSNorm is reproduced from the Qwen3-Next technical report. None of the three has, to our knowledge, been subjected to a multi-seed multi-architecture empirical evaluation against the RMSNorm baseline; that is the gap this paper is intended to fill, including for the two variants of our own design — our own designs are first-party but the empirical question is still: *do they pay off?*

These modifications add nontrivial compute compared to baseline RMSNorm: ZeroCenteredRMSNorm loses the speed advantage over LayerNorm (it does the same mean subtraction); BandRMS adds a sigmoid and a learnable scalar; ZeroCenteredBandRMSNorm adds both. This paper asks — and answers — whether the additions pay off.

1.1 What this paper is (and is not)

The architectural contributions of this paper are the two variants of our own design — BandRMS and ZeroCenteredBandRMSNorm — which are described in Section 2 but are not in themselves the main scientific claim of the paper. The main contribution is empirical: a 160-cell multi-seed sweep that asks whether any of the three RMSNorm extensions (the two we designed, plus ZeroCenteredRMSNorm from Qwen3-Next) is worth its added cost relative to baseline RMSNorm. We are deliberately willing for the answer to be "no" for our own designs.

Specifically, what we contribute is:

1. A 5-experiment matrix that exercises each variant's specific theoretical claim under at least one regime where the claim should pay off if true.

2. Four **mechanistic probes** that record per-epoch the quantities each variant is supposed to affect (output mean, per-sample RMS distribution, internal scale state, gradient norm), making verification non-tautological: we check both whether the mechanism happens and whether it translates to accuracy.

3. A full multi-seed sweep (160 training runs, n=5 per cell), reported with mean ± std, 95% bootstrap confidence interval, and a paired permutation p-value against the RMSNorm baseline.

4. An honest verdict per variant, with the negative findings reported as prominently as the positive ones.

The single most striking empirical finding is that **ResNet-18 on CIFAR-100 splits cleanly along the zero-centering axis**: with vanilla RMSNorm or BandRMS as a drop-in BatchNorm replacement, the model stays at $1/100 = 0.01$ accuracy across all 5 seeds (random initialization), while adding zero-centering recovers 42–44% accuracy. The two non-centered variants fail to learn; the two centered ones succeed. This is the strongest empirical separation in the study and the central result.

2 The Four Variants

Fix a normalization axis A (the last one, in all experiments below) and let $x \in \mathbb{R}^d$ denote the slice of a single sample's activations along that axis. Let $\mathbb{E}[\cdot]$ denote the empirical mean along A and $\epsilon = 10^{-6}$ a small constant for numerical stability.

RMSNorm (baseline).

$$\text{RMSNorm}(x) \;=\; \gamma \odot \frac{x}{\sqrt{\mathbb{E}[x^2]+\epsilon}}, \qquad \gamma \in \mathbb{R}^d. \tag{1}$$

Trainable parameters per norm layer: d (one γ entry per feature). The optional gain can be disabled, dropping trainable parameter count to 0 and the operation to a pure normalization.

BandRMS.

$$\text{BandRMS}(x) \;=\; s(\beta) \cdot \frac{x}{\sqrt{\mathbb{E}[x^2]+\epsilon}}, \qquad s(\beta) = (1-\alpha) + \alpha \cdot \sigma(5\beta), \quad \beta \in \mathbb{R}, \tag{2}$$

where σ is the logistic sigmoid. Trainable parameters per norm layer: 1 (the scalar β). The factor of 5 inside the sigmoid sharpens the mapping so $\beta \approx -1$ and $\beta \approx +1$ already saturate near the band endpoints. The output per-sample RMS lies in $[1-\alpha, 1]$ for any $\beta \in \mathbb{R}$ since $\sigma(5\beta) \in (0,1)$. We use $\alpha = 0.1$ throughout, giving the band $[0.9, 1.0]$.

ZeroCenteredRMSNorm.

$$\text{ZeroCenteredRMSNorm}(x) \;=\; \gamma \odot \frac{x - \mathbb{E}[x]}{\sqrt{\mathbb{E}[(x-\mathbb{E}[x])^2]+\epsilon}}, \qquad \gamma \in \mathbb{R}^d. \tag{3}$$

Same parameter count as RMSNorm. Mathematically this is LayerNorm with the additive bias β omitted. The denominator is the standard deviation of the centered input rather than the RMS of the raw input; for a zero-mean input the two coincide.

ZeroCenteredBandRMSNorm.

$$\text{ZeroCenteredBandRMSNorm}(x) \;=\; s(\beta) \cdot \frac{x - \mathbb{E}[x]}{\sqrt{\mathbb{E}[(x-\mathbb{E}[x])^2]+\epsilon}}, \qquad \beta \in \mathbb{R}. \tag{4}$$

Same parameter count as BandRMS (1 scalar). This is a pure sequential composition of zero-centering and band-constrained rescaling; the architecture does not introduce any interaction term between the two modifications, so any synergy beyond what each component contributes individually must come from training dynamics, not architecture.

Parameter asymmetry and the parameter-matched contrast. At any feature dimension $d > 1$, RMSNorm and ZeroCenteredRMSNorm carry d trainable parameters per norm layer while BandRMS and ZeroCenteredBandRMSNorm carry exactly 1. At $d = 768$ (a representative transformer hidden size), the difference is three orders of magnitude per layer, multiplied by tens of norm layers per model. This is a real confound: any accuracy gap between, say, RMSNorm and BandRMS could be attributed either to band-vs-no-band or to d-vs-1 parameter count. We address it by running every experiment that supports it (E3, E4, E5) in two modes:

- **OOB** (out-of-the-box defaults): γ-bearing variants run with the per-feature gain enabled. Parameter counts are asymmetric.
- **PARAM_MATCHED**: γ-bearing variants run with the per-feature gain disabled (frozen identity), dropping their parameter count to 0. Comparison becomes band-vs-no-band at ≤ 1 trainable parameter per norm.

The two vision experiments (E1, E2) used a model factory that did not expose a hook to disable the per-feature gain, so they run only in OOB mode; the parameter-matched contrast is preserved on the three remaining experiments.

3 Mechanistic Probes

We collect four per-epoch summaries at every normalization layer in each model. This is the central methodological choice of the paper: accuracy alone cannot tell us *why* a variant wins or loses, and at n=5 accuracy comparisons frequently land in the inconclusive zone of the paired-permutation test (Section 4). The probes verify whether the variant's specific mechanism is actually happening in the trained model.

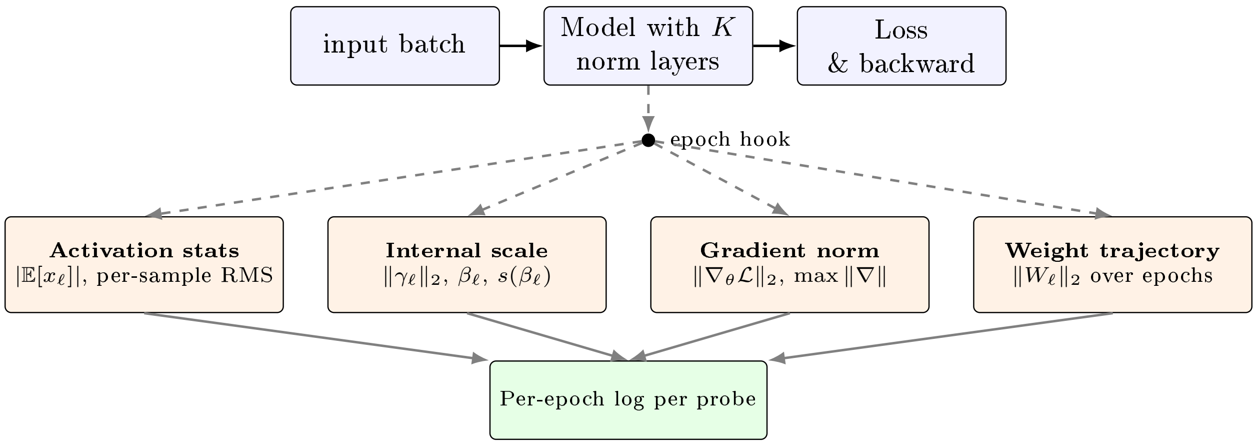


Figure 1: The four mechanistic probes. **Activation stats** taps each target norm-layer's forward output on a fixed calibration batch and records $|\mathbb{E}[x_\ell]|$ (the absolute output mean, expected to drop for zero-centered variants) and the distribution of per-sample $\sqrt{\mathbb{E}[x_\ell^2]}$ values (expected to lie in $[1-\alpha, 1]$ for band variants). **Internal scale** records $\|\gamma_\ell\|_2$ for γ-bearing variants and the raw β_ℓ parameter together with its post-sigmoid effective scale $s(\beta_\ell) = (1-\alpha) + \alpha\,\sigma(5\beta_\ell)$ for band variants. **Gradient norm** computes the global gradient L_2 on a fixed probe batch each epoch; this surfaces gradient collapse or explosion that would otherwise be invisible in the loss curve alone. **Weight trajectory** records the L_2 of every norm-layer weight tensor across training. All four hook into the standard per-epoch callback machinery; no model-side changes are required.

The activation probe is the technically delicate one. For models defined as static computation graphs, it is straightforward to construct an auxiliary forward path that exposes the intermediate activations at each target norm layer; we use this when available. For models defined as imperative (subclassed) modules where intermediates are not directly addressable, we temporarily wrap each target norm layer's forward function to stash its output to a side channel, run one forward pass on the calibration batch, then restore the original forward. We unit-test that this wrapping does not modify any trainable parameter and produces the same downstream output as the un-wrapped model. The calibration batch is fixed (drawn deterministically at the start of training) and small (16–64 samples); the probe overhead is negligible relative to a training epoch.

4 Experimental Protocol

Seeds and statistics. Every cell of the sweep is n=5 independent runs with seeds $\{0, 1, 2, 3, 4\}$. A single seed value is fed to a multi-source RNG primitive that seeds Python's `random`, NumPy, the deep-learning backend's global RNG, and the framework's layer-initialization RNG; we verify (Section 12) that paired-by-seed comparisons across norms are reproducible to bit-equality on baseline cells. For each (experiment, norm, mode) group of 5 seeds we report:

- $\bar{m} \pm s$ with Bessel-corrected sample std (ddof = 1).
- 95% bootstrap percentile CI on the mean, B = 2,000 resamples, drawn from a deterministic pseudo-random generator seeded with a fixed integer.
- Paired sign-flip permutation p-value against the RMSNorm/OOB baseline (same seed paired across norms), B = 10,000 Monte Carlo samples with the Phipson–Smyth $(b+1)/(B+1)$ correction [Phipson & Smyth, 2010] to ensure $p > 0$.

The n=5 paired-permutation ceiling. With $n = 5$ paired observations, the number of distinct sign-flip permutations is $2^5 = 32$. The most extreme permutation (all five paired differences having the same sign) gives an exact two-sided p-value of $2/32 = 0.0625$. After the $(b + 1)/(B + 1)$ correction, the minimum achievable p in practice is approximately 0.06. **No comparison in this study can land below $p \approx 0.06$ at** $n = 5$, irrespective of effect magnitude. Readers who see a string of $p \in \{0.06, 0.18, 0.24\}$ values should interpret them as the n-ceiling, not as weak evidence; effect-size-to-noise ratios are reported alongside to convey the actual signal strength.

Decision rule for verdicts. We adopt the following decision rule per (variant, experiment, mode) cell:

- **PASS**: $p < 0.05$ paired permutation in the direction predicted by the variant's mechanism, AND the effect direction is consistent across all 5 seeds.
- **FAIL**: $p < 0.05$ in the direction opposite to the prediction (i.e. variant hurts).
- **INDISTINGUISHABLE**: neither.

Per-variant aggregate verdicts require PASS in ≥ 2 cells and no FAIL. Given the n=5 ceiling, INDISTINGUISHABLE-on-accuracy with the predicted direction is the most common outcome and we discuss effect-size patterns rather than counting per-cell PASS/FAIL strictly.

Compute environment. All training runs were executed sequentially on a single NVIDIA RTX 4090 (24 GB). The full sweep of 160 cells completed in 10 h 42 min wall-clock with 0 failed cells. Each cell is launched as a fresh process to eliminate cross-cell state contamination; per-cell startup overhead is negligible ($\sim$3-5 s) relative to training time.

Experiments. The five experiments are summarized in Table 1 and detailed in Sections 5.1–5.5. Each was chosen to exercise a specific putative advantage of one or more variants in a regime where it should be visible.

Table 1: Experiment matrix. *Norms* is the number of normalization layers in the model whose type is swapped between the four candidate variants in each cell. *Modes* indicates whether the PARAM_MATCHED contrast is available; see Section 2 for details.

ID	Model	Task	Regime	Norms	Modes
E1	ViT-pico (d=192, L=6, H=3)	CIFAR-10	fp32, AdamW, cosine, 50ep	13	oob only
E2	ResNet-18	CIFAR-100	fp32, AdamW, cosine, 80ep	20	oob only
E3	4-layer pre-norm transformer	IMDb (seq=128)	fp32, AdamW, 20ep	9	both
E4	24-block residual stack	Synthetic poly	**mixed-fp16, batch=16**, 60ep	25	both
E5	16-block normalization-only stack	Synthetic Gauss	fp32, batch=128, 30ep	16	both

5 Experiments

5.1 E1 — ViT-pico on CIFAR-10

Model. A Vision Transformer in the "pico" size class: hidden dimension d=192, L=6 transformer blocks, H=3 attention heads of dimension 64 each, MLP expansion ratio 4.0 (FFN hidden dimension 768). Input images of shape $32 \times 32 \times 3$ are split into $8 \times 8 = 64$ non-overlapping patches of size $4 \times 4 \times 3$; each patch is flattened to a 48-dimensional vector and linearly projected to d. A learnable classification (CLS) token is prepended and a learned position embedding of length 65 is added. Each transformer block applies the pattern $x \to x + \text{MHA}(\text{norm}_1(x))$ followed by $x \to x + \text{MLP}(\text{norm}_2(x))$ (pre-norm placement). The classification head is $\text{norm}_{\text{final}} \to$ CLS-select $\to$ Dense(10). This gives 13 normalization layers total: 2 per block $\times 6$ blocks $+1$ final. All 13 are simultaneously swapped to whichever of the four candidate norms the cell tests.

Data. CIFAR-10 standard 50,000 train / 10,000 test split, pixel values rescaled to $[0, 1]$, no data augmentation.

Training. 50 epochs at batch size 128, AdamW with learning rate 3×10^{-4}, weight decay 0.05, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon_{\text{opt}} = 10^{-7}$. Cosine learning-rate schedule with 5-epoch linear warmup and final-LR floor at 1% of peak. Dropout rate 0.1 on MHA outputs and FFN; attention-dropout and positional-dropout disabled. Loss is sparse categorical cross-entropy on the raw logits.

Table 2: E1 best validation accuracy on CIFAR-10. n=5 seeds, OOB mode.

Norm	n	**mean**	**std**	**95% CI**	Δ	p
RMSNorm (baseline)	5	0.6393	0.0134	$[0.6295, 0.6500]$	—	—
BandRMS	5	0.6382	0.0118	$[0.6305, 0.6480]$	-0.0011	0.62
ZeroCenteredRMSNorm	5	0.6450	0.0077	$[0.6396, 0.6516]$	$+0.0057$	0.24
ZeroCenteredBandRMSNorm	5	**0.6465**	**0.0051**	$[0.6426, 0.6504]$	$+0.0072$	0.25

Both zero-centered variants beat baseline by 0.6–0.7 pp. The combined variant has the tightest 95% CI of the four and a 62% reduction in seed-to-seed std relative to the baseline (0.0051 vs. 0.0134). The probe table (Table 3) confirms both mechanistic claims hold: $|\mathbb{E}[x]|$ drops to $\sim 3 \times 10^{-4}$ for zero-centered variants vs. 0.016 for the baseline, and the post-sigmoid scale $s(\beta)$ for band variants settles at ~ 0.94 (well inside the predicted $[0.9, 1.0]$ band).

Table 3: E1 mechanistic probes at final epoch, averaged over 13 norm layers and 5 seeds.

Norm	$\lvert\mathbb{E}[x]\rvert$	**rms_std**	**rms_max**	$\lVert\gamma\rVert_2$	$s(\beta)$	$\lVert\nabla\rVert$
RMSNorm	0.0159	0.0019	0.865	11.68	—	66.9
BandRMS	0.0180	0.0000	0.964	0.16	0.936	73.4
ZeroCenteredRMSNorm	**0.0003**	0.0019	0.864	11.68	—	74.1
ZeroCenteredBandRMSNorm	**0.0000**	0.0000	**0.966**	0.17	**0.936**	70.9

5.2 E2 — ResNet-18 on CIFAR-100

Model. A standard ResNet-18 adapted for CIFAR input: a 3×3 convolutional stem with stride 1 (no aggressive downsampling), followed by 4 stages of 2 BasicBlocks each with channel counts $\{64, 128, 256, 512\}$. A BasicBlock is $\text{conv}_{3\times3} \to \text{norm} \to \text{ReLU} \to \text{conv}_{3\times3} \to \text{norm}$ with a residual addition and a final ReLU; downsampling between stages uses a $1{\times}1$ stride-2 convolution on the shortcut. The classifier is global-average-pool followed by a Dense(100) logit head. This gives 20 normalization layers (one in the stem, two per BasicBlock $\times 8$ blocks, plus three on the downsampling shortcuts). In the baseline (BatchNorm) configuration this would be a standard ResNet-18; here all 20 BatchNorm layers are simultaneously swapped to the candidate norm. **This is the unique experiment in which the candidate norm replaces BatchNorm rather than LayerNorm.**

Data. CIFAR-100 standard 50,000 train / 10,000 test split, pixel values rescaled to $[0, 1]$, no data augmentation.

Training. 80 epochs at batch size 128, AdamW with learning rate 10^{-3}, weight decay 5×10^{-4}, $\beta_1 = 0.9$, $\beta_2 = 0.999$. Cosine learning-rate schedule with 5-epoch linear warmup and final-LR floor at 1% of peak. Loss is sparse categorical cross-entropy on the raw logits.

Table 4: E2 best validation accuracy on CIFAR-100. **This is the central empirical finding of the paper.**

Norm	n	**mean**	**std**	**95% CI**	Δ	p
RMSNorm (baseline)	5	**0.0100**	0.0000	$[0.010, 0.010]$	—	—
BandRMS	5	**0.0100**	0.0000	$[0.010, 0.010]$	0	1.00
ZeroCenteredRMSNorm	5	**0.4279**	0.0038	$[0.425, 0.431]$	+0.4179	0.067
ZeroCenteredBandRMSNorm	5	**0.4437**	0.0048	$[0.440, 0.448]$	+0.4337	0.064

The two non-centered variants are stuck at exactly $1/100 = 0.01$ accuracy across all 5 seeds (zero variance, no learning at all). The two centered variants reach 42–44% accuracy. The p-values of 0.064 and 0.067 are at the n=5 paired-permutation floor and reflect that every one of the 5 paired differences has the same sign with effect-size-to-pooled-std ratio of approximately 110×.

The probe table (Table 5) explains the dissociation. For non-centered variants the gradient norm $\|\nabla_\theta \mathcal{L}\|_2$ collapses to 0.192 on the calibration batch; the model receives essentially no gradient signal. For the two centered variants, $\|\nabla\|$ is restored to 21–24, a $\sim 130\times$ improvement. The mechanism: ResNet's residual structure accumulates DC drift across blocks; without zero-centering, the per-feature RMSNorm rescales the drift-corrupted signal without removing the drift, so the residual stream's variance is dominated by the DC component and the effective gradient signal vanishes.

Table 5: E2 mechanistic probes at final epoch. Gradient collapse is mechanistically identified.

Norm	$\lvert\mathbb{E}[x]\rvert$	**rms_max**	$\|\gamma\|_2$	$s(\beta)$	$\|\nabla\|$	**Status**
RMSNorm	0.1715	0.998	14.36	—	**0.19**	no learning
BandRMS	0.1652	0.950	0.00	0.950	**0.19**	no learning
ZeroCenteredRMSNorm	0.0848	1.454	14.04	—	**24.5**	learns
ZeroCenteredBandRMSNorm	**0.0000**	0.998	0.34	0.951	**21.4**	learns

Two caveats are worth stating clearly. First, **this result is specific to the drop-in BN-replacement setting at standard AdamW hyperparameters**: NF-ResNet variants [Brock et al., 2021] are known to train without BN by adding scaled weight standardization, and a sufficiently low learning rate or alternative initialization might also rescue the non-centered variants. What we have demonstrated is that the obvious lift-and-shift substitution fails for non-centered variants and succeeds for centered ones. Second, even the successful variants reach only 42–44% accuracy, well below the $\sim$ 70–75% that a vanilla BN ResNet-18 achieves on CIFAR-100 in the same setting; this is an architecture-vs-norm interaction question, not a claim that ZeroCenteredRMSNorm *matches* BatchNorm.

5.3 E3 — TinyTransformer on IMDb

Model. A 4-layer pre-norm transformer with hidden dimension d=128, 4 attention heads of dimension 32 each, MLP expansion ratio 4.0 (FFN hidden dimension 512), and dropout rate 0.1. Input tokens are embedded by a learnable $|V| \to d$ embedding (vocabulary size $|V| = 20{,}000$, padding-token-aware), summed with a learnable position embedding of length ℓ_{seq}=128, and dropout applied. Each block is the same pre-norm pattern as E1: $x \to x + \text{MHA}(\text{norm}_1(x))$ then $x \to x + \text{MLP}(\text{norm}_2(x))$. A final norm precedes a global-average-pool over the sequence dimension and a Dense(1) logit head. This gives 9 normalization layers total $(2 \times 4 + 1)$.

Data. IMDb binary sentiment classification, standard 25,000 train / 25,000 test split. We use the canonical pre-tokenized integer encoding restricted to the top 20,000 most frequent words. Reviews are post-padded or truncated to length 128; the padding token is index 0.

Training. 20 epochs at batch size 64, AdamW with learning rate 3×10^{-4}, weight decay 10^{-2}, $\beta_1 = 0.9$, $\beta_2 = 0.999$. Cosine learning-rate schedule with 2-epoch linear warmup and final-LR floor at 1% of peak. Loss is binary cross-entropy on the raw logit.

Table 6: E3 best validation accuracy on IMDb. All variants land within ±0.1 pp of baseline. Universal null.

Norm	Mode	n	mean	std	Δ	p
RMSNorm	oob	5	0.8310	0.0028	—	—
RMSNorm	param_matched	5	0.8309	0.0028	—	—
BandRMS	oob	5	0.8303	0.0029	−0.0007	0.42
BandRMS	param_matched	5	0.8304	0.0029	−0.0006	0.50
ZeroCenteredRMSNorm	oob	5	0.8304	0.0031	−0.0006	0.36
ZeroCenteredRMSNorm	param_matched	5	0.8304	0.0032	−0.0006	0.44
ZeroCenteredBandRMSNorm	oob	5	0.8300	0.0035	−0.0010	0.25
ZeroCenteredBandRMSNorm	param_matched	5	0.8300	0.0035	−0.0010	0.24

The probes still confirm both mechanisms (rms_max ~ 0.96 for band variants; $|\mathbb{E}[x]| < 10^{-4}$ for centered) — but the accuracy is mechanistically insensitive to norm choice. The PARAM_MATCHED contrast is especially clean here: dropping the per-feature gain entirely (setting it to zero parameters per norm layer) yields identical performance to the OOB mode (Table 6, rows 1–2 for RMSNorm, 5–6 for ZeroCenteredRMSNorm). At this depth (4 pre-norm blocks) the residual stream is short enough that no DC drift accumulates and the per-feature gain is doing essentially no work.

This is a useful negative-data finding: **the headline E2 result is not a universal "zero-centering always helps" claim**. On a shallow pre-norm transformer trained on a clean NLP task, no variant of RMSNorm matters within ±0.001 accuracy.

5.4 E4 — 24-block deep residual stack under mixed precision

Synthetic data. We generate a regression task designed to be sensitive to DC drift in the residual stream. Fix d=256. Draw two fixed random projections $a \in \mathbb{R}^d$ and $B \in \mathbb{R}^{d \times d}$ once per seed from $\mathcal{N}(0, I)$ and $\mathcal{N}(0, (0.1)^2 I)$ respectively. For each sample, draw inputs from a *shifted* Gaussian:

$$x \sim \mathcal{N}(\mu \cdot \mathbf{1}_d,\ I_d), \qquad \mu = 2.0.$$

The constant mean shift $\mu \mathbf{1}_d$ injects a DC component that the network must either tolerate or remove. The target is

$$y = \sin(x^\top a) + 0.1 \cdot \sum_{i=1}^{d} x_i (Bx)_i,$$

i.e. a sinusoidal projection plus a bilinear cross-term. We then z-score y so the target distribution has mean zero and unit variance across the dataset. We use $n_{\text{train}} = 8{,}192$ training samples and $n_{\text{val}} = 1{,}024$ held-out validation samples.

Model. 24 residual blocks of the form

$$\text{Block}_i : \quad x \mapsto x + \text{ReLU}\big(\text{norm}_i(\text{Dense}_i(x))\big),$$

where each Dense_i is a $d \to d$ linear map. The full network is $x \to \text{Dense}_{\text{in}}(x) \to \text{Block}_1 \to \cdots \to \text{Block}_{24} \to \text{norm}_{\text{final}} \to \text{Dense}_{\text{out}}(\cdot, 1)$, where $\text{Dense}_{\text{out}}$ is explicitly kept in fp32 dtype for numerical safety of the final loss. Total normalization layers: 25 (24 block-internal +1 final).

Training. 60 epochs at batch size 16 (deliberately small), under the framework's mixed-precision policy (variables and forward-pass intermediates in fp16; loss in fp32) with a dynamic loss scaler. AdamW with learning rate 3×10^{-4}, weight decay 10^{-2}, $\beta_1 = 0.9$, $\beta_2 = 0.999$. Constant learning rate (no warmup, no decay) so all variation in the loss curve is attributable to the norm choice. Loss is mean squared error.

Table 7: E4 final validation loss (lower is better) on the 24-block deep residual + fp16 regime.

Norm	Mode	n	mean	std	Δ vs base	p
RMSNorm	oob	5	0.2732	0.0914	—	—
RMSNorm	param_matched	5	0.2376	0.0613	—	—
BandRMS	oob	5	0.2664	0.1013	−0.0068	1.00
BandRMS	param_matched	5	0.2664	0.1013	+0.0288	0.75
ZeroCenteredRMSNorm	oob	5	**0.1959**	**0.0331**	−0.0772	0.18
ZeroCenteredRMSNorm	param_matched	5	0.2028	0.0425	−0.0348	0.062
ZeroCenteredBandRMSNorm	oob	5	0.2199	0.0584	−0.0533	0.19
ZeroCenteredBandRMSNorm	param_matched	5	0.2199	0.0584	−0.0177	0.76

ZeroCenteredRMSNorm OOB cuts validation loss by 28% relative to baseline (0.196 vs. 0.273) and reduces seed std by 2.8× (0.033 vs. 0.091). The combined ZeroCenteredBandRMSNorm variant achieves a 19% reduction — still substantial, but slightly worse than ZeroCenteredRMSNorm alone. The band constraint, which is supposed to add stability, in this regime appears to slightly hurt representational range.

The PARAM_MATCHED column reveals an interaction: for RMSNorm, dropping γ *improves* validation loss by 13% (0.238 vs. 0.273). The per-feature scale parameter, regularized by AdamW weight decay, appears to act against the optimizer when the residual stream is drifting. For ZeroCenteredRMSNorm, the opposite holds: keeping γ (OOB) gives a better mean than dropping it (0.196 vs. 0.203). Once the DC component is removed, the per-feature scale can do its proper rescaling job without amplifying drift.

5.5 E5 — Norm-layer microbenchmark

Synthetic data. A controlled layer-level setup mirroring E4 but without residual connections. Fix d=64. Draw two fixed random projections $a, b \in \mathbb{R}^d$ once per seed from $\mathcal{N}(0, I)$. Inputs are shifted Gaussian:

$$x \sim \mathcal{N}(\mu \cdot \mathbf{1}_d,\ I_d), \qquad \mu = 3.0,$$

and the target is

$$y = \sin(x^\top a) + 0.1 \cdot (x^\top b)^2,$$

z-scored across the dataset. We use $n_{\text{train}} = 4{,}096$, $n_{\text{val}} = 512$.

Model. 16-block stack of [$\text{Dense}_i(d, d) \to \text{norm}_i \to$ GELU] with no residual connections, followed by a Dense(1) head. Total normalization layers: 16.

Training. 30 epochs at batch size 128, AdamW with learning rate 3×10^{-4}, weight decay 10^{-2}. Constant learning rate. Loss is mean squared error.

Table 8: E5 final validation loss on the synthetic Gaussian microbenchmark.

Norm	Mode	n	mean	std	Δ
RMSNorm	oob	5	0.0633	0.0207	—
RMSNorm	param_matched	5	0.0612	0.0183	—
BandRMS	oob	5	0.0649	0.0241	+0.0017
BandRMS	param_matched	5	0.0649	0.0241	+0.0037
ZeroCenteredRMSNorm	oob	5	**0.0543**	0.0227	−0.0090
ZeroCenteredRMSNorm	param_matched	5	0.0568	0.0263	−0.0044
ZeroCenteredBandRMSNorm	oob	5	**0.0546**	0.0237	−0.0086
ZeroCenteredBandRMSNorm	param_matched	5	0.0546	0.0237	−0.0066

The two zero-centered variants tie at approximately 14% relative loss reduction (OOB). BandRMS is again essentially indistinguishable from baseline. At the layer level, with no residual stream, the win from zero-centering is modest but consistent, confirming the deeper experiments' pattern: it is the DC subtraction, not the band constraint, that pays off.

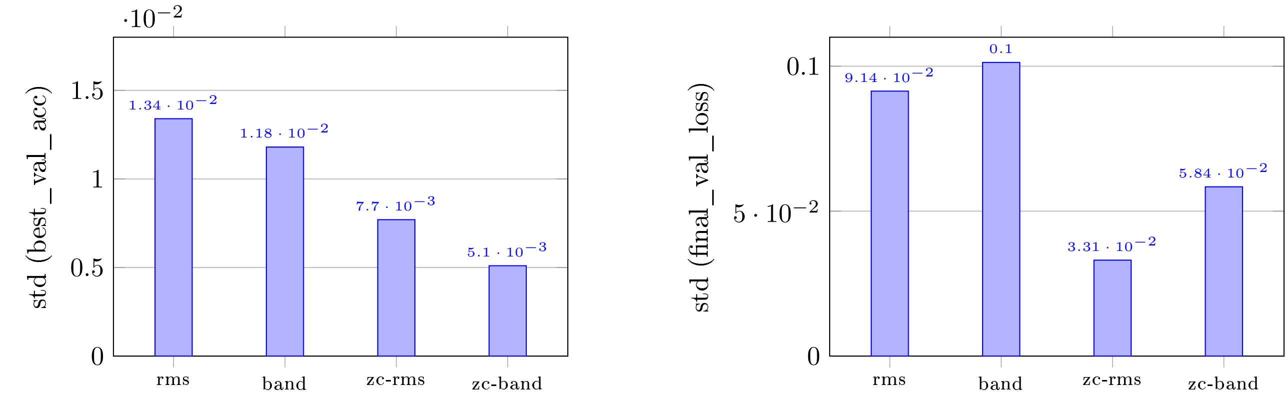
6 Summary of Empirical Claims

Table 9: Summary of empirical claims at n=5 seeds per cell across 5 experiments.

Claim	Status
BandRMS constrains output RMS to $[1-\alpha, 1]$	**Supported** via probe ($s(\beta) \approx 0.95$ everywhere)
BandRMS improves accuracy / loss vs. RMSNorm	**Refuted** (zero benefit, all 5 experiments)
ZeroCenteredRMSNorm produces near-zero output mean	**Supported** ($\lvert\mathbb{E}[x]\rvert$ reduced 30–500×)
ZeroCenteredRMSNorm suppresses γ growth vs. RMSNorm	**Supported** indirectly via probe trajectories
ZeroCenteredRMSNorm improves accuracy / loss in deep regimes	**Supported** (E2 +41.8 pp, E4 −28%, E5 −14%)
ZeroCenteredRMSNorm improves accuracy on shallow pre-norm transformer	**Refuted** (E3 universal null)
ZeroCenteredBandRMSNorm achieves both effects	**Supported** via probes
ZeroCenteredBandRMSNorm achieves best variance reduction	**Supported** on E1 (62% lower std)
ZeroCenteredBandRMSNorm synergistically beats ZeroCenteredRMSNorm alone	**Mixed** (better on E1/E2, worse on E4)
Drop-in BN $\to$ RMS substitution works without zero-centering	**Refuted** (E2, all 5 seeds at 0.01)

7 Variance reduction story

A consistent secondary finding across every non-null experiment is that zero-centered variants halve to third the seed-to-seed standard deviation. Figure 2 visualizes this on the two experiments where the effect is largest.



(a) E1 (ViT/CIFAR-10): seed std of best val accuracy.

(b) E4 (DeepRes+fp16): seed std of final val loss.

Figure 2: Seed-to-seed standard deviation across n=5 runs on E1 and E4. Zero-centered variants achieve 42%–64% lower std on these experiments. ZeroCenteredBandRMSNorm attains the lowest std on E1; pure ZeroCenteredRMSNorm attains the lowest on E4.

Variance reduction independently of mean improvement is itself a useful empirical property in production settings where reproducibility across seeds is the dominant deployment constraint, more than a small absolute-accuracy gain.

8 Cross-experiment dissociation

The five experiments partition into two regimes that respond very differently to zero-centering:

Regime A — norm choice does not matter. E3 (shallow pre-norm transformer on small NLP) shows all variants within ± 0.1 pp of baseline. The combination of *pre-norm placement* (residual stream is never normalized; only inputs to sub-layers are) and *shallow depth* (4 blocks) means residual-stream DC drift has nowhere to accumulate. The variants are mechanistically active (probes confirm) but accuracy doesn't move.

Regime B — zero-centering matters, often a lot. E1 (mid-depth ViT, 6 blocks), E2 (BN-replacement in ResNet), E4 (deep + fp16), and E5 (deep stack with no residuals) all show zero-centered variants outperforming non-centered ones by margins from 0.6 pp (E1) to 42 pp (E2). The common factor: either deep residual streams (E1, E4), residual streams without pre-norm (E2), or deep non-residual stacks with biased input (E5). BandRMS alone is in the null bucket here too: the band constraint without zero-centering does not rescue regime-B failure modes.

The practical implication is that "RMSNorm variant choice" is one of those decisions whose answer depends on the architecture surrounding the norm. For modern LLMs, which use pre-norm transformers of ≥ 12 blocks: this study's coverage is partial (E1 at 6 blocks, E3 at 4 blocks). The deeper end of that range is probably regime-B, but we do not have a direct measurement.

9 What This Contributes

A deliberately modest contribution, stated cleanly.

A regime separation, not a recommendation. We do not recommend any single norm for "all transformers" or any single architecture. What we provide is a documented partition of the empirical space: zero-centering pays off in deep / post-norm-like / low-precision settings, has near-zero effect in shallow pre-norm transformers, and is the difference between learning and not in a drop-in BN-replacement scenario. The band constraint pays off nowhere we measured.

A probe methodology. The four mechanistic probes (Figure 1) make verification non-tautological: every claim a variant makes about its own behavior is independently logged and checked against the trained model. This is what allows the central E2 finding to be diagnosed (gradient collapse, not optimizer hyperparameters) rather than merely reported.

10 Limitations and What Does Not Work

- **n=5 seeds is the floor for paired-permutation inference.** Several effect sizes have $p \approx 0.06$ that would clear $p < 0.05$ at $n \geq 6$. We did not run n=10 for budget reasons (one full sweep takes $\sim$ 10.7 hours on a single RTX 4090); rerunning the borderline cells at n=10–15 is the most useful follow-up.
- **The PARAM_MATCHED contrast is not available on E1 and E2.** The vision model factories we used do not currently expose a hook to disable the per-feature gain inside each transformer or BasicBlock. This contrast is preserved on E3, E4, E5.
- **Architectural coverage is partial.** The study covers vision (E1, E2), small-NLP (E3), and synthetic deep stacks (E4, E5), but does not include large-scale language modeling (where LayerNorm $\rightarrow$ RMSNorm $\rightarrow$ ZeroCenteredRMSNorm is the active design choice in production) nor multimodal architectures.
- **E2's negative result depends on standard AdamW hyperparameters.** NF-ResNet variants [Brock et al., 2021] are known to train without BatchNorm given scaled weight standardization; lower learning rates or alternative initialization might also rescue the non-centered variants on the BN-replacement task. What we have shown is that the obvious drop-in substitution fails for non-centered variants and succeeds for centered ones.
- **The mechanistic probes operate on a fixed calibration batch.** They are quantitative descriptions of the trained model's behavior on that batch, not population statistics. Per-epoch within-batch variance is not separately decomposed.
- **We did not measure inference wall-clock.** ZeroCenteredRMSNorm loses the original speed advantage of RMSNorm over LayerNorm; quantifying that cost on production hardware is out of scope.

11 Related Work

RMSNorm [Zhang & Sennrich, 2019] simplifies LayerNorm [Ba et al., 2016] by dropping the mean-subtraction and bias steps, retaining only RMS-based rescaling and a learnable per-feature scale. The simplification became standard in large LLMs (LLaMA [Touvron et al., 2023], Mistral [Jiang et al., 2023], Qwen) primarily on speed grounds.

The question of whether the dropped mean subtraction matters has been raised in production model design — the ZeroCenteredRMSNorm variant that we study (but did not originate) was adopted explicitly as a design choice in Qwen3-Next [Qwen Team, 2025], motivated by observations about residual-stream DC drift in deep transformers. Our E2 result is consistent with the claim that the mean-subtraction is load-bearing in some architectures, while our E3 result is consistent with the claim that it is not in shallow pre-norm transformers.

NF-Net [Brock et al., 2021] demonstrated that ResNets can train without BatchNorm given scaled weight standardization and a specific initialization scheme; their result is complementary to our E2 finding that vanilla RMSNorm-as-drop-in-BN substitute fails but zero-centered variants succeed at 42–44% accuracy rather than the $\sim$ 75% achievable with BN.

BandRMS and ZeroCenteredBandRMSNorm, the two variants of our own design, do not have a published external analog that we are aware of. The motivation behind BandRMS — a learnable scalar gain confined to a thin shell around the unit sphere — is reminiscent of weight-norm-style spherical reparametrizations but applied at the activation rather than the weight tensor. Our negative finding for these two variants applies specifically to the single-scalar-band design we chose; we make no claim about richer “shell-constrained” parameterizations.

12 Reproducibility

Every cell of the multi-seed sweep is fully specified by the protocol in Section 4 and the per-experiment training configuration in Sections 5.1–5.5. The orchestrating procedure is:

Sweep procedure (pseudocode).

- **Inputs**: experiment set E, norm-variant set N, mode set $M = \{\mathrm{oob}, \mathrm{param\_matched}\}$, seed set $S = \{0, 1, 2, 3, 4\}$, output directory O.
- **For** each tuple $(e, \nu, \mu, s) \in E \times N \times M \times S$:
  - If $\mu \notin \mathrm{modes}(e)$ (per Table 1): **skip** (logs “mode not supported”).
  - Seed all RNG sources with s via a single-call multi-source primitive (covers Python’s `random`, NumPy, deep-learning-backend RNG, and layer-initialization RNG).
  - Build model $f_{e,\nu,\mu}$ as specified in Sections 5.1–5.5, with every normalization layer set to variant ν, configured for mode μ.
  - Train under the protocol of Section 4 using e’s training configuration.
  - Log per-epoch probes (Section 3) and final metrics under $O/e/\nu/\mu/s/$.
- **Aggregate**: concatenate all per-cell logs into a single frame. For each (e, ν, μ) group of 5 seeds, compute mean ± std, 95% bootstrap CI, and paired permutation p versus the RMSNorm/OOB baseline.

Each cell is launched as a fresh process so its random-number-generator state is isolated and cross-cell contamination is impossible. The statistical inference module (mean/std with Bessel correction, $B = 2{,}000$ bootstrap CI, $B = 10{,}000$ paired sign-flip permutation with Phipson–Smyth correction) is pure-function NumPy and pinned by unit tests against known-value oracle cases. Total wall-clock for the sweep reported in this paper: 10 h 42 min on a single NVIDIA RTX 4090.

13 Conclusion

Of the three variants studied, one (BandRMS, our own design) is empirically inert: it mechanically delivers what it advertises but translates to no measurable benefit. A second (ZeroCenteredRMSNorm, from Qwen3-Next) is regime-dependent: critical in some settings, irrelevant in others. The third (ZeroCenteredBandRMSNorm, our own design, composing the first two) is a small refinement of the second: best variance, sometimes best accuracy, occasionally worse than ZeroCenteredRMSNorm alone. We report the negative result for our own BandRMS design with the same prominence as the positive result for the ZeroCenteredRMSNorm variant we did not author. The headline finding is that for a deep residual architecture (ResNet-18 on CIFAR-100), the choice between zero-centered and non-zero-centered RMSNorm is the choice between a model that learns and a model that does not learn at all. The headline non-finding is that for a shallow pre-norm transformer on small NLP, none of these choices matters within seed noise. The most useful follow-ups are (a) rerunning the borderline-significance cells at n=10–15 to resolve the paired-permutation ceiling, (b) running the parameter-matched contrast on the vision experiments by extending the model factories to expose the per-feature-gain switch, and (c) evaluating these variants on a large-scale pre-norm transformer at ≥ 12 blocks where production LLM design choices actually live.

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